Exercise 2.1

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Given: $y'''(x) + y''(x) + y'(x) + y(x) = 0$; $y(0) = 0$, $y'(0) = 1$, $y''(0) = 2$

a) Transform to first-order-system:

Define a new variable z in the following way: $z_k = y^{(k-1)}$

$$\Rightarrow \quad z = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} y \\ y' \\ y'' \end{pmatrix} \quad \Rightarrow \quad \dot{z} = \begin{pmatrix} y' \\ y'' \\ -(y + y' + y'') \end{pmatrix} = \begin{pmatrix} z_2 \\ z_3 \\ -(z_1 + z_2 + z_3) \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{pmatrix} z$$

$$\Rightarrow \quad \dot{z} = f(x, z(x)) \; ; \quad z \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

(Just note that I chose the notation z^i to not cause confusion with part a) *. z^i still stands for the i.-step in the Euler-method)

b) Two steps of explicit Euler-method: $z^{i+1} = z^i + h f(x_i, \underbrace{z(x_i)}_{z^i}) = z^i + h A z^i$

step size: $h = \frac{1}{2}$

$$z^0 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}; \quad z^1 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \frac{1}{2} f\left(\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, z\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0{,}5 \\ 1 \\ -1{,}5 \end{pmatrix} = \begin{pmatrix} 0{,}5 \\ 2 \\ 0{,}5 \end{pmatrix}$$

$$z^2 = \begin{pmatrix} 0{,}5 \\ 1 \\ 0{,}5 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{pmatrix}\begin{pmatrix} 0{,}5 \\ 2 \\ 0{,}5 \end{pmatrix} = \begin{pmatrix} 0{,}5 \\ 1 \\ 0{,}5 \end{pmatrix} + \begin{pmatrix} 1 \\ 0{,}25 \\ -1{,}5 \end{pmatrix} = \begin{pmatrix} 1{,}5 \\ 1{,}25 \\ -1 \end{pmatrix}$$

$$\Rightarrow \quad y\left(\tfrac{1}{2}\right) = z_1^1 = 0{,}5 \; ; \quad y'\left(\tfrac{1}{2}\right) = z_2^1 = 2 \; ; \quad y''\left(\tfrac{1}{2}\right) = z_3^1 = 0{,}5$$

$$y(1) = z_2^1 = 1{,}5 \; ; \quad y'(1) = z_2^2 = 1{,}25 \; ; \quad y''(1) = z_3^2 = -1$$

c) $y(x) = -\cos(x) + 2\sin(x) + \exp(-x)$; $y'(x) = \sin(x) + 2\cos(x) - \exp(-x)$; $y''(x) = \cos(x) - 2\sin(x) + \exp(-x)$

Analytic solution	Euler	Comparison
$y(\frac{1}{2}) = -\cos(\frac{1}{2}) + 2\sin(\frac{1}{2}) + e^{-\frac{1}{2}} \approx 0{,}6878$	0,5	Absolute difference between 0,0105 and 0,6958
$y(1) = -\cos(1) + 2\sin(1) + e^{-1} \approx 1{,}5105$	2	↳ not very accurate, but if we consider the content of the lecture, we know that the error scales with h. $h = \frac{1}{2}$ is quite a big step size, so we had to expect an "error-range" like that.
$y'(\frac{1}{2}) = \sin(\frac{1}{2}) + 2\cos(\frac{1}{2}) - e^{-\frac{1}{2}} \approx 1{,}6281$	2	
$y'(1) = \sin(1) + 2\cos(1) - e^{-1} \approx 1{,}5542$	2,25	
$y''(\frac{1}{2}) = \cos(\frac{1}{2}) - 2\sin(\frac{1}{2}) + e^{-\frac{1}{2}} \approx 0{,}5253$	0,5	
$y''(1) = \cos(1) - 2\sin(1) + e^{-1} \approx -0{,}1748$	-1	

Exercise 2.4

a) $\dot{x}(t) = y(t)$; $x(0) = 1$

$\dot{y}(t) = -x(t)$; $y(0) = 0$

Set $z = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \dot{z} = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} y(t) \\ -x(t) \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}_{=:A} z$; $z \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\Rightarrow$ Implicit Euler: $z_{i+1} = z_i + h f(t_{i+1}, z_{i+1}(t)) = z_i + hAz_{i+1}$

$\Rightarrow z_{i+1} = \begin{pmatrix} x_{i+1} \\ y_{i+1} \end{pmatrix} = \begin{pmatrix} x_i \\ y_i \end{pmatrix} + h \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x_{i+1} \\ y_{i+1} \end{pmatrix}$

$\Leftrightarrow \begin{pmatrix} x_i \\ y_i \end{pmatrix} = (I - hA) \begin{pmatrix} x_{i+1} \\ y_{i+1} \end{pmatrix}$

$\overset{(*)}{\Leftrightarrow} \underbrace{(I - hA)^{-1}}_{=:C} \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} x_{i+1} \\ y_{i+1} \end{pmatrix}$

$\Leftrightarrow C \cdot \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} x_{i+1} \\ y_{i+1} \end{pmatrix}$

(*) $I - hA = \begin{pmatrix} 1 & -h \\ h & 1 \end{pmatrix}$; $\det(I - hA) = 1 + h^2 \neq 0 \quad \forall h \in \mathbb{R}$

$\Rightarrow$ Invers exists: $\left(\begin{array}{cc|cc} 1 & -h & 1 & 0 \\ h & 1 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cc|cc} 1 & -h & 1 & 0 \\ 0 & 1+h^2 & -h & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cc|cc} 1 & -h & 1 & 0 \\ 0 & 1 & \frac{-h}{1+h^2} & \frac{1}{1+h^2} \end{array}\right)$

$\longrightarrow \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{1+h^2} & \frac{h}{1+h^2} \\ 0 & 1 & \frac{-h}{1+h^2} & \frac{1}{1+h^2} \end{array}\right)$

$\Rightarrow$ Invers: $\frac{1}{1+h^2} \begin{pmatrix} 1 & h \\ -h & 1 \end{pmatrix} =: C$

spectral radius: $\rho(C) = \max_i |\lambda_i(C)|$

eigenvalues:

$\Rightarrow \det(tI - C) = \det \begin{pmatrix} t - \frac{1}{1+h^2} & -\frac{h}{1+h^2} \\ \frac{h}{1+h^2} & t - \frac{1}{1+h^2} \end{pmatrix} = \left(t - \frac{1}{1+h^2}\right)^2 + \left(\frac{h}{1+h^2}\right)^2$

$= t + \frac{2t}{1+h^2} + \frac{1}{(1+h^2)^2} + \frac{h^2}{(1+h^2)^2}$

$= t + \frac{2t}{1+h^2} + \frac{1+h^2}{(1+h^2)^2}$

$= t + \frac{2t}{1+h^2} + \frac{1}{1+h^2}$

abc: $t_{1/2} = \frac{-\frac{2}{1+h^2} \pm \sqrt{\left(\frac{2}{1+h^2}\right)^2 - 4 \cdot 1 \cdot \frac{1}{1+h^2}}}{2}$

$\vdots$

$t_{1/2} = \frac{1 \pm ih}{1+h^2}$

$\Rightarrow \quad |\lambda_{1/2}| = \left|\frac{1 \pm ih}{1+h^2}\right| = \sqrt{\left(\frac{1}{1+h^2}\right)^2 + \left(\frac{h}{1+h^2}\right)^2} = \sqrt{\frac{1+h^2}{(1+h^2)^2}} = \sqrt{\frac{1}{1+h^2}} = \frac{1}{\sqrt{1+h^2}}$

Since $h > 0$ it follows that $\rho(C) = \frac{1}{\sqrt{1+h^2}} < 1$

↳ C describes a continuous linear function and $\rho(C) < 1$

$\Rightarrow (I - C)^{-1}$ exists, because $\sum_{n=1}^{\infty} C^n = (I - C)^{-1}$ converges (Neumann-Reihe)

↳ $(I - C)^{-1} = (I - I + hA) = (hA)^{-1} = h^{-1} A^{-1}$.

Since $\sum_{n=1}^{\infty} C^n$ converges $\Rightarrow \|C^n\| \xrightarrow{n \to \infty} 0$ in matrix norm. Since $\begin{bmatrix} x_n \\ y_n \end{bmatrix} = C^n \cdot \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$

$$0 \leq \left\|\begin{bmatrix} x_n \\ y_n \end{bmatrix}\right\|_2 = \left\| C^n \cdot \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \right\| \leq \|C^n\| \left\|\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}\right\| \quad \text{| matrix norm inequality}$$

as $n \to \infty$ $\left\|\begin{bmatrix} x_n \\ y_n \end{bmatrix}\right\|_2 \to 0 \Leftrightarrow \begin{bmatrix} x_n \\ y_n \end{bmatrix} \to \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

So approximations converge to zero, i.e., method is asymptotically stable.